Fuel Pathway
Code
Lower Calorific
Value (LCV,
MJ/g)
share in fuel blend
(%MJ(LCV) / MJ(LCV))
WtT GHG emission
factor (GWP100,
gCO2eq/MJ(LCV))
+
Part B-1 (Part B-2)15
Emissions credits related to biogenic carbon
source (, in gCO2/g fuel based on
GWP100)
Emissions credits related to source of captured
carbon (, in gCO2/g fuel based on GWP100)
+
Part C-1 Part C-2 Part C-3
Value 1 (carbon source
NOT taken into account):
TtW GHG emission factor
(GWP100, gCO2eq/MJ(LCV))
Value 2 (carbon source
taken into account):
TtW GHG emission factor
(GWP100, gCO2eq/MJ(LCV))
Energy Converter
+
Part D Part E
WtW GHG emission factor
(GWP100, gCO2eq/MJ(LCV))
Note: Part D = Part A-5 + Part C-2
Sustainability (Certification)16

8.3 Different parties (fuel suppliers, owners/operators, Administration/RO, etc.) may use different parts of the FLL for different purposes along the fuel pathway. As such, each interested party may use those parts of the FLL as relevant to their activities and purposes

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SOURCE: MEPC.376(80).pdf
different parts of the FLL for different purposes along the fuel pathway. As such, each interested party may use those parts of the FLL as relevant to their activities and purposes
rather than the complete, integrated document.

8.4 The five main parts of the FLL are explained below.

.1 Part A of the FLL indicates:

.1 fuel type (Part A-1);

.2 fuel pathway code (Part A-2);

.3 lower calorific value (Part A-3, in MJ/g); and

.4 WtT GHG emission factor (Part A-5, in gCO2eq/MJ(LCV) calculated on GWP100).

Part A-4 is only applicable when a fuel batch is supplied to the ship as a blend of fuels with different fuel pathway code (hereinafter referred to as the "fuel blend") and indicates the share of each blend component in the fuel

15 Pending further methodological guidance to be developed by the Organization (see section 5).

16 Pending further guidance to be developed by the Organization.

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SOURCE: MEPC.376(80).pdf

RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 19

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blend (in %MJ (LCV)/MJ(LCV)). If fuel blends are denoted on volume -basis, a re -calculation on energy basis based on the LCV values of the blend components is required;

For the fuel blend supplied to a ship, the information on fuel type for the mixture is presented under Part A -1 on top of its components, named by percentual order of composition in the fuel, e.g. X (70%), Y (20%), Z (10%). Part A-5, Part C-1, Part C-2 and Part D are the average value weighted on energy share (% MJ(LCV)) /MJ(LCV))) of each fuel component, while Part A-2 to A-4, Part B and Part E are kept blank. Each component of the fuel blend with a specific fuel pathways code is presented in a separate ro w below the row for the fuel blend;

.2 Part B of the FLL indicates the carbon credits related to the carbon source, including:

.1 ec (Part B-1, in gCO2 / g fuel calculated on GWP100); (and

.2 eccu (Part B-2, in gCO2/g fuel calculated on GWP100)),17

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SOURCE: MEPC.376(80).pdf

.2 Part B of the FLL indicates the carbon credits related to the carbon source, including:

.1 ec (Part B-1, in gCO2 / g fuel calculated on GWP100); (and

.2 eccu (Part B-2, in gCO2/g fuel calculated on GWP100)),17

as defined in section 5 of these guidelines;

.3 Part C of the FLL indicates the TtW GHG emission factor of the fuel type in conjunction with the energy converter(s) on board the ship (Part C -3). The TtW GHG emission factor of the fuel type is further categorized as:

.1 Value 1 where carbon source is not taken into account (Part C-1, in gCO2eq/MJ(LCV) calculated on GWP100); and

.2 Value 2 where carbon source is taken into account (Part C-2, in gCO2eq/MJ(LCV) calculated on GWP100),

as defined in section 5 of these guidelines;

.4 Part D of the FLL indicates the WtW GHG emission factor of the fuel type (in gCO2eq/MJ(LCV) calculated on GWP100), which is always the sum of Part A-5 and Part C-2; and

.5 Part E of the FLL indicates the sustainability performance of the fuel as per Section 7 of these guidelines.

PART III: DEFAULT EMISSION FACTORS AND ACTUAL VALUES

9 DEFAULT EMISSION FACTORS

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SOURCE: MEPC.376(80).pdf

A-5 and Part C-2; and
.5 Part E of the FLL indicates the sustainability performance of the fuel as per Section 7 of these guidelines.

PART III: DEFAULT EMISSION FACTORS AND ACTUAL VALUES

9 DEFAULT EMISSION FACTORS

9.1 The principles and the procedure described for the determination of default emission factors under this section 9 have been used for the establishment of default emission factors and should remain valid for the factors that will be established.

17 Pending further methodological guidance to be developed by the Organization. For more details on the eccu parameter and Part B-2 of the FLL, refer to sections 5.2 and 8.2, respectively.

9.2 WtT default emission factors should be calculated using representative and conservative assumptions, which encompass variable performance of feedstock -fuel pathways across world regions and States.

9.3 To establish a WtT default emission factor, at least three reference values from three

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SOURCE: MEPC.376(80).pdf
conservative assumptions, which encompass variable performance of feedstock -fuel pathways across world regions and States.

9.3 To establish a WtT default emission factor, at least three reference values from three different, representative, sources should be considered. Among the three (or more) values considered, the upper emission value should be selected as default, and the range of available emission factors should be provided for informative purposes. The reference values should be accompanied by the relevant technical and scientific information (see Template set out in appendix 4) and evaluated against the corresponding information as appropriate, including the agreement between the reference values.

9.4 Emissions related to carbon stock changes caused by direct land-use change (DLUC) (el) and emissions savings from soil carbon accumulation via improved agricultural management (esca) are considered as zero for the establishment of the initial default emission factors. Similarly, this is the case also for the parameters related to carbon capture and storage (ccs), which require further development.

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